

Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models ¹
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Submission Date	

1 Aim and Objective

To implement linear and regularized regression models (Linear, Ridge, Lasso, and Elastic Net) for predicting a continuous target variable – the sanctioned loan amount – evaluate their performance using multiple regression metrics, visualize model behaviour, and analyze overfitting, underfitting, and bias–variance characteristics.

2 Dataset Description

The Kaggle “Predict Loan Amount Data” dataset (the standard Loan Prediction dataset) was used, containing numerical and categorical features related to loan applications, with **LoanAmount** as the continuous target variable to be predicted.

Dataset Name	Loan Amount Prediction (Loan Prediction Dataset)
Dataset Source	Kaggle – <i>Predict Loan Amount Data</i>
Original Number of Samples	614
Samples after Dropping Missing Target	592
Number of Input Features	10 (6 categorical, 4 numerical) – expands to 14 after one-hot encoding
Target Variable	LoanAmount (continuous, range 9–700, mean 146.4, std 85.6)
Target Skewness	2.68 (strongly right-skewed)
Missing Values (Predictors)	Gender (13), Married (2), Dependents (13), Self_Employed (31), Loan_Amount_Term (14), Credit_History (49)
Train-Test Split	80:20 (473 train / 119 test), <code>random_state=42</code>

3 Preprocessing Steps

- **Missing target handling:** 22 rows with a missing **LoanAmount** value were dropped, since imputing the regression target itself would fabricate ground truth rather than genuinely missing predictor information.
- **Missing value handling (predictors):** Categorical columns (Gender, Married, Dependents, Self_Employed) were imputed with their mode; numerical columns (Loan_Amount_Term, Credit_History) were imputed with their median.

- **Encoding categorical variables:** All six categorical features (Gender, Married, Dependents, Education, Self_Employed, Property_Area) plus Loan_Status were one-hot encoded (`drop="first"` to avoid the dummy-variable trap), expanding 10 raw features into 14 encoded features.
- **Standardizing numerical features:** ApplicantIncome, CoapplicantIncome, Loan_Amount_Term and Credit_History were standardized (zero mean, unit variance) using `StandardScaler`, fit only on the training split and applied to both splits via a `ColumnTransformer` inside a `Pipeline` to prevent data leakage.

4 Implementation Details

All four models were implemented using scikit-learn, wrapped in a single `Pipeline` (pre-processing $\rightarrow$ model) to keep preprocessing consistent across cross-validation folds and the held-out test set.

- **Linear Regression:** Trained as the baseline with no regularization and no hyperparameters to tune.
- **Ridge, Lasso, Elastic Net:** Each tuned via `GridSearchCV` with 5-fold cross-validation (`KFold`, `shuffle=True`, `random_state=42`), optimizing for R^2 , over the hyperparameter grids specified in the assignment (Ridge: $\alpha \in \{0.01, 0.1, 1, 10, 100\}$; Lasso: $\alpha \in \{0.001, 0.01, 0.1, 1, 10\}$; Elastic Net: $\alpha \in \{0.01, 0.1, 1, 10\}$, `l1_ratio` $\in \{0.2, 0.5, 0.8\}$).
- All four fitted pipelines were then evaluated identically: 5-fold CV metrics on the training set, and a single held-out evaluation on the untouched 119-row test set.

5 Visualizations

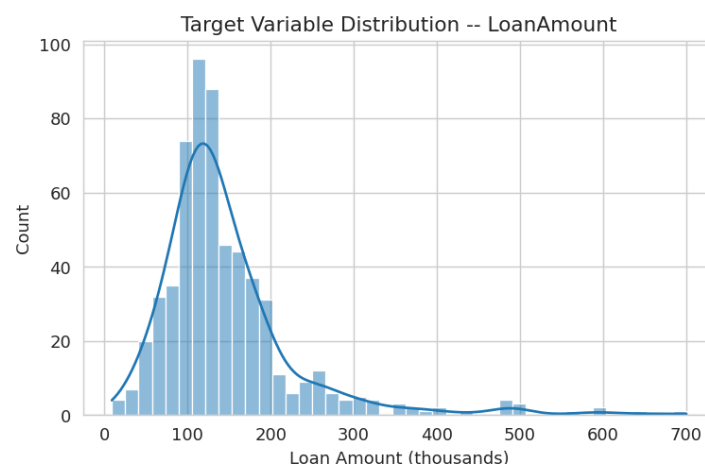


Figure 1: Target variable (LoanAmount) distribution

Inference: LoanAmount is strongly right-skewed (skewness ≈ 2.68), with most sanctioned amounts clustered between 100 and 170 (thousands) and a long tail extending to

700. This skew is a primary driver of the residual pattern seen later, and would justify a log-transform of the target in a follow-up iteration of this experiment.

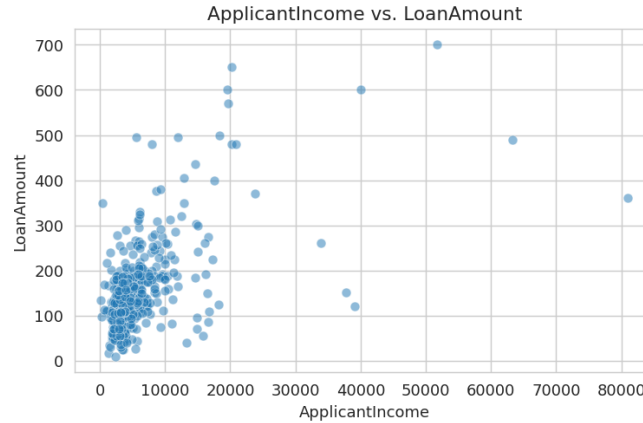


Figure 2: ApplicantIncome vs. LoanAmount

Inference: ApplicantIncome shows a moderate positive correlation with LoanAmount ($r \approx 0.57$), the strongest of any single numeric feature, consistent with it emerging as the most heavily weighted feature in every model’s coefficients (Section 6). The relationship is noisy and heteroscedastic – variance in LoanAmount increases at higher incomes.

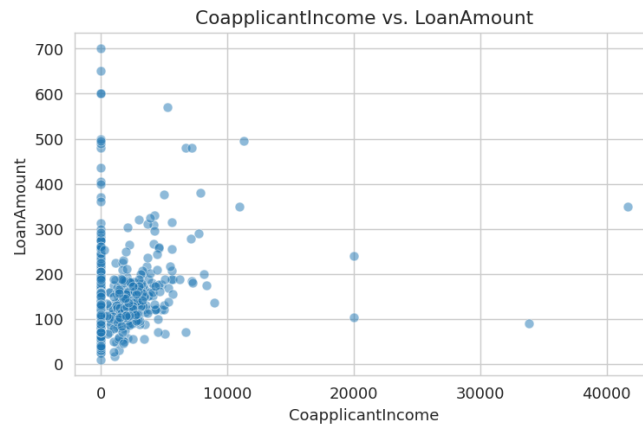


Figure 3: CoapplicantIncome vs. LoanAmount

Inference: CoapplicantIncome is only weakly correlated with LoanAmount ($r \approx 0.19$), and a large cluster of applicants report zero co-applicant income, indicating many single-applicant loans. This weaker relationship is reflected in a consistently smaller coefficient for this feature across all four models.

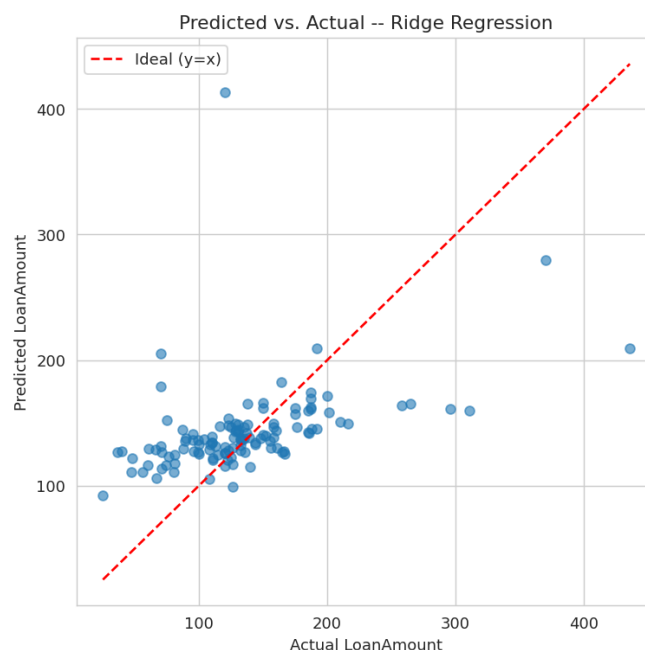


Figure 4: Predicted vs. actual LoanAmount – best model (Ridge Regression)

Inference: Points scatter loosely around the ideal $y = x$ line rather than tracking it tightly, and the model visibly under-predicts for the highest actual loan amounts (the right-skew tail), consistent with a linear model struggling to capture the more extreme, rarer high-value loans.

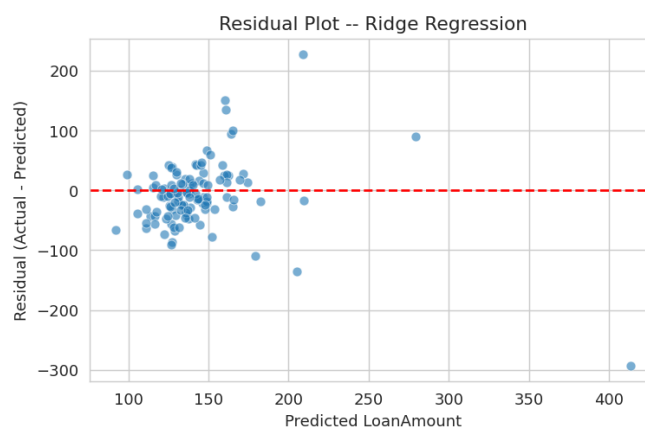


Figure 5: Residual plot – best model (Ridge Regression)

Inference: Residuals are not randomly scattered around zero – they fan out and skew positive as predicted values increase, a classic heteroscedasticity signature tied directly to the right-skewed target distribution seen in Figure 1. This indicates the constant-variance assumption of ordinary least squares is violated, and a log-transformed target would likely produce more homogeneous residuals.

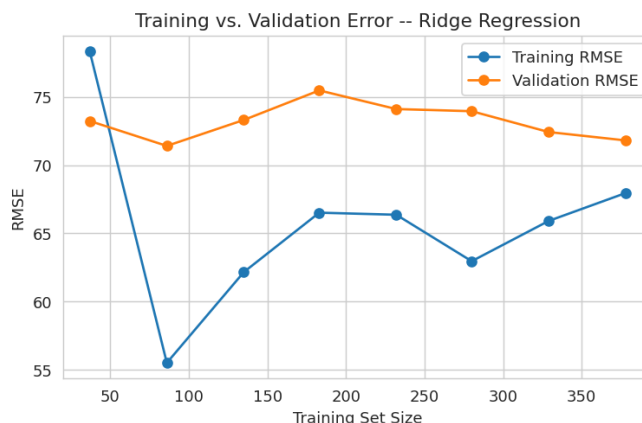


Figure 6: Training error vs. validation error (learning curve) – best model (Ridge Regression)

Inference: Training and validation RMSE converge to a similar, persistently high range (≈ 55 – 75) rather than validation error dropping toward training error as more data is added. This gap-closing-but-high-floor pattern is the signature of **underfitting** / **high bias** rather than overfitting: the linear feature set does not have enough capacity to model the target’s variance, so more data alone will not substantially help.

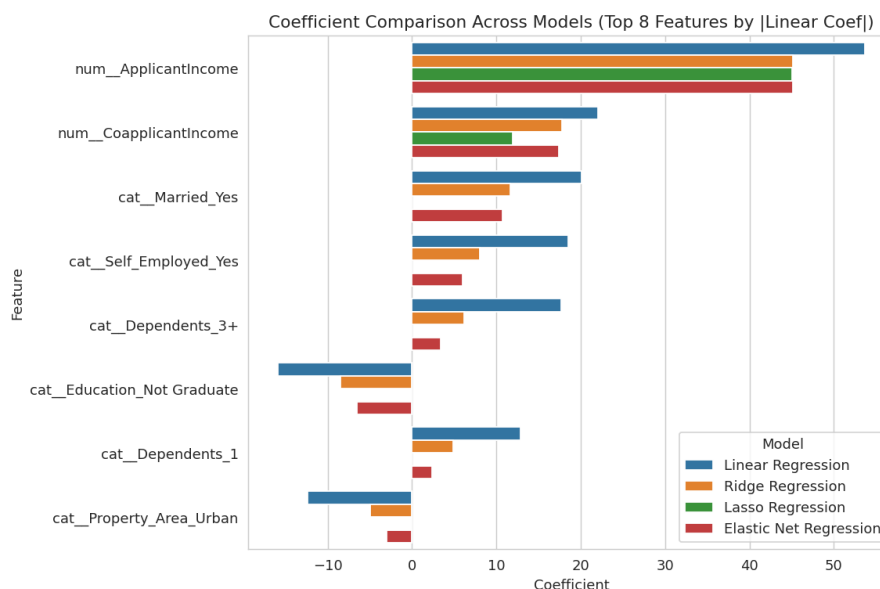


Figure 7: Coefficient comparison across models (top 8 features by $|\text{Linear coefficient}|$)

Inference: Ridge shrinks every coefficient toward zero without eliminating any, Elastic Net shrinks them further while retaining small non-zero weights on most features, and Lasso is the most aggressive – it zeroes out Married_Yes, Self_Employed_Yes, Dependents_3+, Education_Not Graduate, Dependents_1, and Property_Area_Urban entirely, keeping only ApplicantIncome and CoapplicantIncome with non-trivial weight. This is a direct, visible demonstration of Lasso’s feature-selection property versus Ridge’s pure shrinkage.

6 Performance Tables

Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best R^2 CV
Ridge Regression	Grid Search	$\alpha = 100$	0.3403
Lasso Regression	Grid Search	$\alpha = 10$	0.3151
Elastic Net Regression	Grid Search	$\alpha = 1$, $l1_ratio = 0.8$	0.3356

Cross-Validation Performance (K = 5, training set)

Model	MAE	MSE	RMSE	R^2
Linear Regression	47.32	5590.81	74.01	0.3015
Ridge Regression	44.50	5199.02	71.80	0.3403
Lasso Regression	46.45	5435.54	73.31	0.3151
Elastic Net Regression	44.45	5244.73	72.10	0.3356

Test Set Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	36.99	3339.92	57.79	0.0940
Ridge Regression	36.84	3079.74	55.50	0.1646
Lasso Regression	37.80	3181.25	56.40	0.1370
Elastic Net Regression	36.81	3087.84	55.57	0.1624

Training time for all four models was under 10 milliseconds on the 473-row training set (Linear: 7.2ms, Ridge: 6.7ms, Lasso: 7.0ms, Elastic Net: 6.8ms) – negligible and not a practical differentiator at this dataset size.

Note on CV vs. test R^2 : All four models score noticeably higher R^2 on 5-fold cross-validation (≈ 0.30 – 0.34) than on the single held-out test set (≈ 0.09 – 0.16). With only 592 usable samples, a single 119-row test split carries high variance in its R^2 estimate; the CV numbers, averaged over five folds, are the more stable estimate of true generalization performance.

Effect of Regularization on Coefficients (Top Features)

Feature	Linear	Ridge	Lasso	Elastic Net
ApplicantIncome	53.68	45.12	45.10	45.14
CoapplicantIncome	22.06	17.76	11.91	17.43
Married_Yes	20.10	11.68	0.00	10.71

Feature	Linear	Ridge	Lasso	Elastic Net
Self_Employed_Yes	18.51	8.03	0.00	5.98
Dependents_3+	17.64	6.20	0.00	3.43
Education_Not Graduate	-15.91	-8.47	-0.00	-6.52

7 Overfitting and Underfitting Analysis

- **Training vs. validation error:** The learning curve (Figure 6) shows training RMSE (55–78) and validation RMSE (71–75) are already close at small training sizes and stay close as more data is added – there is no widening gap that would indicate overfitting.
- **Effect of regularization strength:** Grid search selected a large $\alpha = 100$ for Ridge, a relatively strong $\alpha = 10$ for Lasso, and $\alpha = 1$ with a high l1.ratio of 0.8 for Elastic Net (closer to pure Lasso behaviour). The fact that stronger regularization was consistently preferred over weaker settings is itself evidence that the baseline linear model was not egregiously overfitting – if it were, weak regularization would have already generalized poorly and the search would have pushed toward much larger α to compensate for high variance; instead the improvement from regularization is modest (CV R^2 0.30 $\rightarrow$ 0.34), consistent with a bias-dominated, not variance-dominated, problem.
- **Improvement in generalization after tuning:** Ridge and Elastic Net improved test RMSE from 57.79 (Linear) to about 55.5, and test R^2 nearly doubled (0.094 $\rightarrow$ 0.16–0.16). This is a genuine but modest improvement – regularization helped, but the ceiling on all four models’ performance is set by the limited linear signal available in the features, not by uncontrolled overfitting.

Conclusion of this section: The dominant issue for this dataset is **underfitting (high bias)**, not overfitting. All four models plateau at a similar, modest R^2 , and adding regularization only marginally improves results – indicating that the ten available features do not fully explain the variance in LoanAmount, likely because larger, more influential drivers of loan amount (e.g. property value, requested amount, lender policy) are not present in this feature set.

8 Bias–Variance Analysis

- **Bias behaviour of Linear Regression:** Linear Regression has the lowest bias of the four models (it fits the training data most closely, as seen in its lowest CV RMSE being only marginally above the regularized models), but this experiment shows even its unconstrained fit cannot exceed $R^2 \approx 0.30$ on training folds – indicating the model family itself (a linear function of these 14 encoded features) has a bias ceiling well below what the target’s variance requires.
- **Variance reduction using Ridge and Elastic Net:** Ridge achieved the best CV R^2 (0.3403) and the best test RMSE (55.50) of all four models, and Elastic

Net was close behind – both shrink coefficient magnitudes (visible directly in Figure 8/Section 6’s coefficient table) which reduces sensitivity to noise in individual training folds, producing the small but consistent generalization gain seen between the Linear Regression and Ridge rows of the CV table.

- **Feature sparsity effect in Lasso:** Lasso zeroed out 6 of the 14 encoded features entirely (Married_Yes, Self_Employed_Yes, Dependents_3+, Education_Not_Graduate, Dependents_1, Property_Area_Urban), retaining only ApplicantIncome and CoapplicantIncome with substantial weight. This aggressive sparsity slightly *increased* bias relative to Ridge (Lasso’s CV R^2 of 0.3151 is the second-lowest of the four models), suggesting some of the zeroed categorical features did carry a small amount of genuine signal that Ridge’s softer shrinkage preserved.

9 Observations and Conclusion

All four models converge to a similar, modest performance ceiling (test R^2 between 0.09 and 0.16), with **Ridge Regression** emerging as the best-performing model on both cross-validation ($R^2 = 0.3403$) and the held-out test set (RMSE = 55.50, $R^2 = 0.1646$), tuned to $\alpha = 100$. Lasso’s aggressive feature elimination did not improve performance over Ridge or Elastic Net, indicating most of the encoded categorical features carry at least weak signal rather than being pure noise. The overall pattern across every diagnostic (learning curve, CV-vs-test gap, coefficient shrinkage) points to **underfitting driven by limited feature information** rather than overfitting: ApplicantIncome and CoapplicantIncome dominate every model’s coefficients, and the available categorical features (marital status, education, employment, property area) contribute only marginal additional explanatory power. A meaningful next step would be engineering additional features (e.g. income-to-loan-term ratio) or acquiring features more directly tied to the lender’s actual approval formula, rather than further regularization tuning, since regularization strength is not the binding constraint here.

The trade-off between accuracy and model complexity in this experiment is therefore small: Ridge’s mild additional complexity (one tuned hyperparameter) bought a real but modest gain over the simpler, parameter-free Linear Regression baseline, while Lasso’s added complexity (automatic feature selection) did not translate into a performance advantage on this dataset.

10 References

- [1] Scikit-learn Developers, “Linear Models,” https://scikit-learn.org/stable/modules/linear_model.html
- [2] Scikit-learn Developers, “Tuning the hyper-parameters of an estimator,” https://scikit-learn.org/stable/modules/grid_search.html
- [3] Analytics Vidhya / Kaggle, “Predict Loan Amount Data,” <https://www.kaggle.com/datasets>